

State Fiscal and Regulatory Policy Effects on Economic Performance

Evidence from U.S. state-level panel data, 2000-2020

Tyson Test

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DATA

50 states
2000–2020



SAMPLE

Panel data
N = 1,050



METHOD

State fixed-effects
regressions



OUTCOMES

Real GSP per capita
Real private-sector
GSP per capita



RESEARCH DESIGN



Question

How do state fiscal and regulatory policies relate to state economic performance across U.S. states?



Sources

Cato Freedom in the 50 States;
BEA; Census state population estimates



Panel structure

50 states × 21 years (2000–2020)



POLICY VARIABLES

- State government consumption and investment
- Tax burden measures
- Public-sector employment
- Occupational licensing indicators
- Regulatory takings and land-use regulation



MAIN FINDING



Fiscal variables were more consistently associated with state economic performance than most regulatory variables tested. Several regulatory variables were not statistically significant in the fixed-effects models.



PORTFOLIO SIGNAL

Applied econometric research • Public finance • Panel-data workflow • Policy interpretation • Conference-style presentation



Selected undergraduate research project summary graphic.

Executive Summary

- This project examines how fiscal and regulatory policy variables relate to economic performance across U.S. states from 2000 to 2020.
- The analysis uses a state-level panel and fixed-effects regression models to compare associations with real gross state product (GSP) per capita and real private-sector GSP per capita.
- Fiscal variables were more consistently associated with economic performance than most regulatory variables tested.
- The project served as an early applied econometrics research paper prior to my honors thesis and was presented at the Missouri Valley Economics Association in Kansas City in Fall 2024.

Project at a Glance

Research question	How do state fiscal and regulatory policies relate to state economic performance across U.S. states?
Data	50 states over 21 years, 2000-2020 (N = 1,050 observations)
Outcomes	Real GSP per capita and real private-sector GSP per capita
Method	State fixed-effects regressions using fiscal and regulatory policy variables
Main takeaway	Fiscal policy variables were more consistently associated with economic performance than most regulatory variables tested.

1. Introduction

Understanding how public policies relate to economic output is important for lawmakers, researchers, and voters. Because the United States is a federal system, states differ in taxation, spending, regulation, and institutional design. This variation makes the states useful “laboratories of democracy” for studying public finance questions.

This project asks whether state fiscal and regulatory policy differences are associated with differences in economic performance. The analysis focuses on both total real GSP per capita and real private-sector GSP per capita, allowing the project to distinguish broad economic output from output attributable to private-sector activity.

2. Data and Variables

The policy variables come from the Cato Institute’s Freedom in the 50 States dataset. The original dataset includes a wide range of fiscal, regulatory, and personal-freedom measures. This project uses a subset of fiscal and regulatory variables that are relevant to state economic performance.

Economic output measures come from the Bureau of Economic Analysis (BEA). Population data come from Census population estimates. Real GSP and real private-sector GSP are divided by state population to produce per-capita outcome variables.

Policy variables considered

- State government consumption and investment relative to income
- Local and state tax burden measures
- State and local public employment relative to private-sector employment
- State debt relative to income
- Land-use regulation, eminent domain reform, and regulatory-compensation requirements
- Minimum-to-median wage ratio, discrimination-law coverage, health-insurance regulation, telecom deregulation, occupational licensing, and other regulatory-policy measures

3. Empirical Strategy

The empirical strategy estimates fixed-effects regression models using state-level panel data. Fixed effects are appropriate because states may consistently differ in economic performance for reasons not fully captured by the policy variables, such as geography, industrial mix, climate, or historical development patterns.

Simplified model: $\text{Economic performance}_{it} = \text{policy variables}_{it} + \text{state fixed effects} + \text{error}_{it}$

The two primary outcomes are real GSP per capita and real private-sector GSP per capita. Standard errors are reported in parentheses in the results table. The coefficients should be interpreted as conditional associations within the project’s empirical design, not as definitive causal estimates.

4. Results

The main results suggest that several fiscal policy variables are associated with economic performance in the fixed-effects models. In particular, higher state government consumption and investment relative to income and higher state tax burden are associated with lower per-capita economic output measures in the models. Several regulatory variables are not statistically significant, and many have smaller coefficient magnitudes.

Selected fixed-effects regression results

Variable	Real GSP per capita	Real private-sector GSP per capita
State government consumption & investment / income	-230.6* (99.75)	-265.7* (128.3)
State taxation / income	-500.4** (175.2)	-617.0** (225.3)
Sum fiscal policy variables	-54.83** (21.19)	-67.51* (27.25)
Local taxation / income	267.7*** (74.13)	343.0*** (95.36)
Log competing local jurisdictions	8.443*** (2.190)	10.44*** (2.817)
N	1,050	1,050
R-squared	0.452	0.513

Notes: Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Table reports selected variables from the project's fixed-effects regressions.

5. Interpretation and Limitations

The project's central finding is that fiscal policy variables appear more consistently associated with state economic performance than most regulatory variables tested. This does not imply that regulation is irrelevant; rather, the selected regulatory variables in this model often show weaker or less consistent relationships with the outcome variables.

Several results should be interpreted cautiously. State-level policy variables can be highly correlated with other institutional, economic, geographic, and political characteristics. Fixed effects help account for persistent state-level differences, but they do not fully solve endogeneity or omitted-variable concerns.

The project also uses a limited subset of policy variables from a broader source dataset. A more extensive study could expand the variable set, refine model specification, add additional controls, and test robustness across alternative time periods and outcome measures.

6. Conclusion

Overall, the project suggests that fiscal-policy differences across states are meaningfully associated with economic performance in the fixed-effects models. The analysis provided early experience with panel-data methods, public-finance interpretation, and conference-style research communication, and it helped build the foundation for later econometric work in my honors thesis.

References

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